

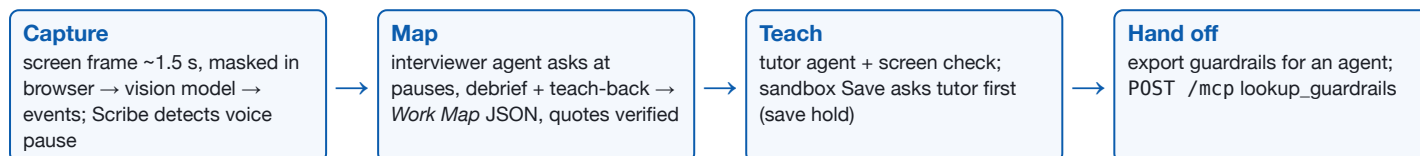
GroundZero: The AI Apprentice

An AI that learns *why* an expert decides, then teaches the next person (or agent) to decide the same way.

1. CHALLENGE TACKLED

Problem: what makes an experienced employee good is never written down. Screen recordings show *what* was clicked, not *why*, and limits ("equipment over 5,000 EUR is always capex") stay invisible until someone breaks them. **Users:** the **expert** (an accounts-payable clerk), the **new hire** who must take over, and a **software agent** that must stop where the expert would.

What we built: three modules on one sandbox ERP with fake invoices. **Capture:** the expert shares a screen and works while talking; a voice agent asks *why* only when the screen is still and the expert is quiet, at least three questions, one about a guardrail. **Map:** spoken debrief with follow-ups, then a teach-back the expert confirms, producing a **Work Map** (every step = decision + reason in the expert's verbatim words + guardrails + screen moment). **Teach:** a voice tutor explains each step, has the new hire predict the next decision, and on an unseen EUR 7,200 invoice stops a wrong save *before* it happens.



2. TOOLS / ML MODELS USED

Tool	Role
ElevenAgents (Expressive)	Interviewer and tutor voice agents
ElevenLabs Scribe v2 Realtime	Live speech-to-text; pause detection
ElevenLabs TTS / STT	Fallback voice when agents are off
Claude Sonnet 5.5	Vision (screen events), Work Map, teach-back, tutor checks, agent-export test, via one function (lib/model.js)
Claude Sonnet 4.5	LLM behind the ElevenAgents agents
tesseract.js (in browser)	Opt-in OCR to black out IBAN / email / phone on frames
Node 20, plain ES modules	No framework or build step; one handler for local and Vercel
Vercel, Postgres, S3	Deploy; saved action trees and videos
Remotion	"How it works" clips on the landing page

4. WHAT WAS CHALLENGING

- When to speak.** Two signals must agree: Scribe says the expert stopped talking and a 128x72 pixel probe says the screen is still (2.5 s / 1.8 s). Early versions asked too many generic questions; now capped at 2 per segment, 5 per 10 min, ranked guardrail first.
- Grounding.** LLMs invent plausible reasons. We reject any quote not found verbatim in the expert's speech.
- "Before it is saved".** Screen checks lag, so the sandbox Save asks the open tutor first and stays unsaved on a violation.
- Privacy.** Regex for speech, region and OCR masking for frames; if masking fails the frame is withheld.
- Environment.** An Anthropic key without a workspace id made every call fail with 400; a misspelled env var silently disabled voice.

HONEST LIMITS AND NEXT STEPS

Not done: no full rehearsal with real voice and a shared screen yet, so pause latency and questions per 10 min are not measured; the sample Work Map is hand-written; the guardrail MCP tool is not registered on the tutor; names and addresses on screen are not detected and the recorded video file is not redacted. **If we had 24 more hours,** we would record a real expert session on the sandbox as the sample, run the two-person rehearsal and publish the measured numbers, and add Presidio-grade redaction.

3. WHAT WORKED WELL

- 157 / 157 tests pass** (node:test) across all three modules.
- Unseen case caught:** live call to the real model flagged cost center 4711 on the unseen EUR 7,200 invoice and quoted the expert's words.
- Agent hand-off:** an agent holding only the exported instructions stopped and asked the controller (no asset number). node scripts/verify-export.js prints PASS.
- No invented reasons:** a quote is kept only if it appears in what the expert said; otherwise it is dropped and counted.
- Voice round trip:** ElevenLabs TTS then Scribe returned the exact sentence.
- Trust:** off-the-record button, masked frames, and the expert can delete any step or answer afterwards; exports no longer contain it.

5. HOW WE SPENT THE BUILD

Clock (Oct 4, from commits)	Work
01:43 – 06:15	Describe pipeline, Work Map, tutor, ElevenAgents and Scribe libs
06:15 – 09:00	Sandbox ERP, debrief, question pacing, compare, guardrail MCP
09:00 – 12:30	Fewer questions, segment tree, QR/AR side demo, intent turns, route-map view
12:30 – 14:00	README restated as requirements, landing page and clips, Vercel deploy

Moonshot: people first, then agents. The same Work Map teaches a new hire today and constrains an AI agent tomorrow. Data: none beyond fake sandbox invoices (public/sandbox/).